



Hospital Engineering and Management

Lecture 8: Operation Theatre Suite – Planning, Engineering Services and Safety

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Course Code: UBM702

Learning Objectives

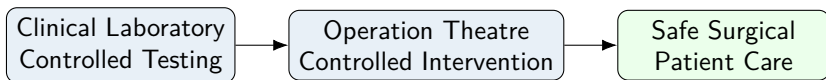
By the end of this lecture, students should be able to:

- ▶ explain the purpose and functional organization of an operation theatre (OT) suite;
- ▶ apply zoning, circulation and adjacency principles to OT planning;
- ▶ identify essential HVAC, electrical, medical-gas and equipment requirements;
- ▶ explain how sterile workflow and infection-control principles influence design;
- ▶ recognize major electrical, fire, gas, equipment and environmental hazards;
- ▶ describe the biomedical engineer's role in commissioning, maintenance and risk management.

Connection with the Previous Lecture

Lecture 7 Established

Clinical laboratories depend on controlled workflow, reliable utilities, equipment, biosafety and traceability to produce trustworthy results.



Key Shift

In the OT, engineering systems do not only support diagnosis – they directly support an invasive procedure on an anaesthetized patient.

Opening Question: What Must Work Before the First Incision?

A surgeon is ready to begin. Which systems must already be functioning correctly?

- ▶ anaesthesia workstation;
- ▶ patient monitor;
- ▶ medical oxygen and suction;
- ▶ surgical lights;
- ▶ electrosurgical unit;
- ▶ sterile instruments.
- ▶ HVAC and pressure control;
- ▶ emergency electrical power;
- ▶ communication and alarms;
- ▶ infection-control process;
- ▶ trained personnel;
- ▶ recovery and emergency backup.

Safe surgery is a system outcome, not the work of one person or one device.

What Is an Operation Theatre Suite?

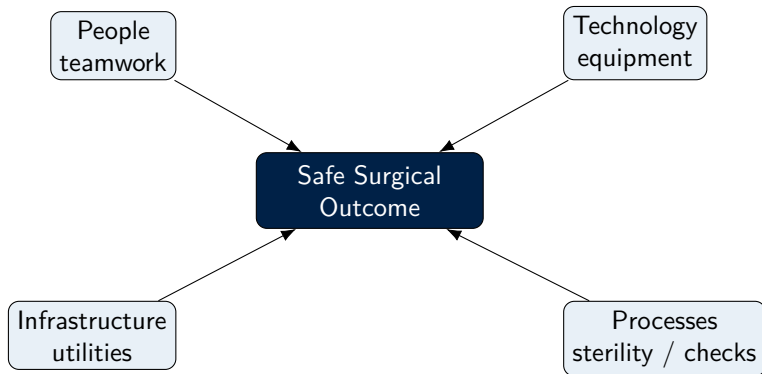
Definition

An operation theatre suite is a controlled hospital environment designed for surgical and interventional procedures, anaesthesia, sterile preparation, patient transfer and immediate postoperative care.

It is more than the operating room itself. A complete suite may include:

- ▶ reception and patient-transfer areas;
- ▶ pre-anaesthesia / preparation areas;
- ▶ operating rooms;
- ▶ scrub and gowning areas;
- ▶ sterile stores and clean utility;
- ▶ dirty utility / disposal pathway;
- ▶ post-anaesthesia care unit (PACU);
- ▶ staff changing, documentation and support spaces.

Why Is the OT a High-Risk Socio-Technical System?



Why Failure Matters

The patient may be unconscious, unstable and dependent on several life-supporting systems at the same time.

Types of Operation Theatres

By Clinical Specialty

- ▶ general surgery;
- ▶ orthopaedic surgery;
- ▶ neurosurgery;
- ▶ cardiac / vascular surgery;
- ▶ obstetric and gynaecologic surgery;
- ▶ ophthalmic / ENT surgery.

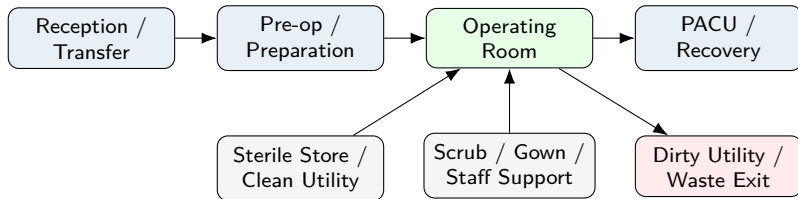
By Technology / Use

- ▶ emergency OT;
- ▶ day-care / ambulatory OT;
- ▶ minimally invasive / laparoscopic OT;
- ▶ robotic OT;
- ▶ hybrid OT with advanced imaging;
- ▶ procedure / interventional room.

Planning Principle

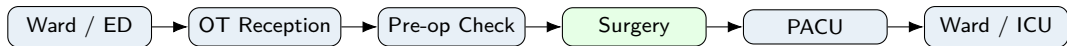
Different procedures create different requirements for room size, equipment, imaging, gases, power, shielding and workflow.

Functional Components of an OT Suite



Planning must support both clinical flow and contamination control.

Patient Journey Through the OT Suite



At every transition, verify:

- ▶ patient identity and procedure;
- ▶ documentation and consent status;
- ▶ allergies and critical risks;
- ▶ required equipment, implants and blood products;
- ▶ continuity of monitoring and handover information.

Core OT Planning Principles

1. **Separation:** keep clean, sterile and contaminated flows from unnecessary crossing.
2. **Control:** restrict access to progressively cleaner zones.
3. **Adjacency:** place strongly interacting functions close together.
4. **Visibility:** support supervision, communication and patient observation.
5. **Reliability:** provide dependable utilities and contingency arrangements.
6. **Maintainability:** allow safe service access without disrupting sterile workflow.
7. **Flexibility:** plan for future equipment, technology and procedure changes.

Engineering View

Good OT design reduces unnecessary movement, contamination opportunities, cable clutter, delays and failure consequences.

Location and Adjacency in the Hospital

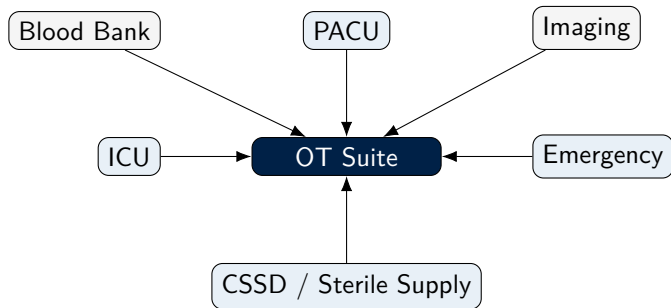
The OT suite should have controlled access and strong functional links with:

- ▶ surgical wards and day-care areas;
- ▶ emergency department for emergency surgery;
- ▶ ICU / critical care for high-risk postoperative patients;
- ▶ blood bank / transfusion services;
- ▶ imaging for selected procedures;
- ▶ CSSD / sterile supply;
- ▶ laboratory and pathology services;
- ▶ biomedical engineering and technical support.

Design Goal

Minimize travel distance for critical flows without turning the OT corridor into a general hospital passage.

Adjacency Map: What Should Be Close?

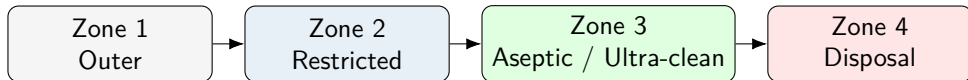


Not Every Adjacency Is Equal

The final layout depends on hospital size, case mix, emergency workload, vertical transport and local architectural constraints.

Concept of Zoning

Zoning controls movement from less-controlled areas toward the most aseptic spaces.



Indian IPC Guidance

NCDC's national infection-prevention guidance describes outer, restricted, aseptic and disposal zones and emphasizes controlled movement between them.

What Happens in Each Zone?

Zone	Typical Function	Examples
Outer	Reception / interface with general hospital	patient reception, administrative functions, waiting, transfer interface
Restricted	Transition into controlled OT environment	changing, clean corridors, staff preparation, selected stores
Aseptic	Highest cleanliness and procedural control	operating room, scrub / gown support, sterile preparation
Disposal	Controlled removal of contaminated material	dirty utility, used instruments, waste exit, soiled-material pathway

Important

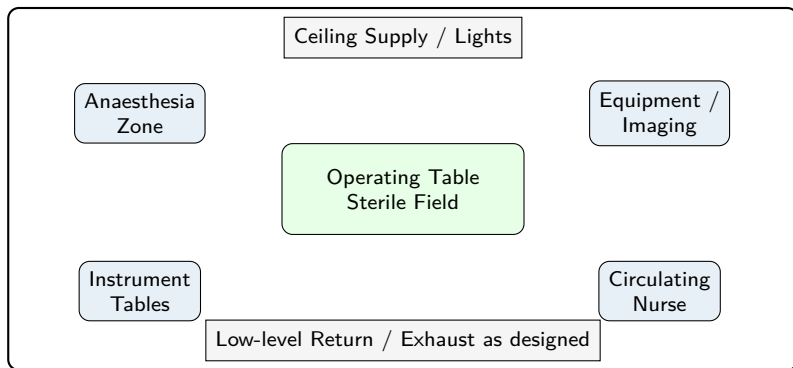
Zoning is useful only when doors, routes, staff behavior, cleaning and pressure relationships support the intended separation.

Clean and Dirty Circulation



- ▶ avoid backtracking of used instruments through clean supply routes;
- ▶ provide a defined route for waste and soiled materials;
- ▶ separate public traffic from controlled OT circulation;
- ▶ use workflow discipline even when architecture cannot provide completely separate corridors.

Functional Layout of an Operating Room



Design Question

Can staff move around the patient without colliding with cables, pendants, imaging equipment, sterile tables or each other?

Operating Table and Team Zones

Around the Table

- ▶ surgeon and assistants;
- ▶ scrub nurse / technologist;
- ▶ anaesthesia team at head end;
- ▶ circulating nurse;
- ▶ imaging / equipment operator when required.

Engineering Implications

- ▶ unobstructed access to airway and patient;
- ▶ safe boom / pendant movement;
- ▶ adequate power and gas outlets;
- ▶ cable and hose management;
- ▶ emergency access from multiple sides.

Room size alone does not guarantee usable space.

Doors, Walls, Floors and Surfaces

OT finishes should support cleaning, durability and infection control.

- ▶ smooth, non-porous and easily cleanable surfaces;
- ▶ sealed joints and minimal ledges that collect dust;
- ▶ coved wall-floor junctions where appropriate;
- ▶ doors that support controlled airflow and equipment movement;
- ▶ impact-resistant surfaces near mobile equipment routes;
- ▶ no unnecessary openings or clutter in the operating room.

NCDC Principle

Cracks, crevices and difficult-to-clean surfaces can harbor contamination; OT finishes should be selected for effective environmental cleaning.

HVAC: What Must the System Achieve?

Clinical / IPC Objectives

- ▶ reduce airborne contamination;
- ▶ maintain directional airflow;
- ▶ control temperature;
- ▶ control humidity as required;
- ▶ remove heat and odors;
- ▶ maintain staff and patient comfort.

Engineering Objectives

- ▶ stable pressure relationships;
- ▶ appropriate air-change rate;
- ▶ suitable filtration;
- ▶ reliable sensing and alarms;
- ▶ accessible maintenance;
- ▶ commissioning and periodic verification.

Pressure Cascade: Cleaner to Less Clean



- ▶ positive pressure helps limit entry of less-clean air when doors are closed;
- ▶ frequent door opening disturbs the intended pressure and airflow pattern;
- ▶ pressure should be measured and trended where required by the adopted standard;
- ▶ alarms need a defined response – not just a display on the wall.

Ventilation Parameters: What Do Guidelines Say?

India's NCDC infection-prevention guidance for OTs describes:

- ▶ positive-pressure airflow from cleaner toward less-clean areas;
- ▶ high-efficiency filtration and regular filter-performance monitoring;
- ▶ a recommended ventilation rate of **20 air changes per hour**;
- ▶ high-level air supply and low-level outlets to promote downward clean-air movement;
- ▶ a temperature range of approximately **18–24°C**.

Do Not Design from One Slide

Final HVAC values must follow the currently adopted local code, accreditation requirement, procedure type, equipment needs and project-specific risk assessment. ASHRAE Standard 170-2025 is one current international ventilation reference for health-care facilities.

HVAC Monitoring and Failure Response

Useful indicators may include:

- ▶ room pressure differential;
- ▶ temperature and humidity;
- ▶ filter differential pressure;
- ▶ fan / AHU status;
- ▶ alarm history and trend data;
- ▶ maintenance and validation records.

When Pressure Is Lost

The response should consider procedure status, duration of failure, door activity, environmental conditions, infection-control risk and whether the room remains suitable for the planned case.

Electrical System in the OT

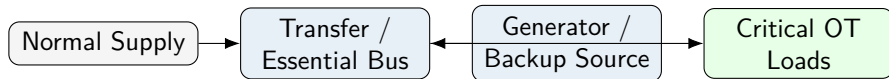
OT electrical design must support many devices simultaneously and safely.

- ▶ adequate number and distribution of socket outlets;
- ▶ essential / emergency circuits for critical equipment;
- ▶ protective earthing and equipotential bonding as required;
- ▶ leakage-current control and electrical-safety testing;
- ▶ safe routing of cables and extension avoidance;
- ▶ dedicated circuits for high-load equipment where required;
- ▶ surge protection / UPS for sensitive systems based on risk.

Patient Risk

The anaesthetized patient may have invasive lines, conductive fluids and multiple connected devices, increasing the importance of electrical integrity.

Emergency Power and Continuity



Critical loads may include:

- ▶ anaesthesia workstation and patient monitoring;
- ▶ selected surgical lights;
- ▶ essential communication / IT;
- ▶ selected equipment needed to complete or safely stop a procedure;
- ▶ critical HVAC controls as defined by the facility design.

Surgical Lighting

A surgical light must provide more than high illumination.

- ▶ focused, shadow-controlled illumination of the surgical field;
- ▶ stable positioning and smooth arm movement;
- ▶ suitable color rendering for tissue differentiation;
- ▶ low heat burden at the field;
- ▶ cleanable handles / sterile control options;
- ▶ emergency-power continuity according to facility design.

Maintenance Checks

Mechanical drift, light intensity, arm movement, handle condition, electrical safety and backup operation should be checked systematically.

Medical Gas and Vacuum Systems

Common OT services may include:

- ▶ oxygen;
- ▶ medical air;
- ▶ vacuum / suction;
- ▶ nitrous oxide where used;
- ▶ carbon dioxide for selected procedures / insufflation;
- ▶ anaesthetic-gas scavenging where required.

Safety Requirements

Correct gas identity, pressure, terminal labeling, alarm functionality, source redundancy, zone-valve access and preventive maintenance are essential.

Anaesthesia Workstation: A Life-Support System

Major Functions

- ▶ deliver medical gases;
- ▶ deliver volatile anaesthetic agents where used;
- ▶ provide ventilation;
- ▶ support breathing circuits;
- ▶ monitor gas / ventilation variables;
- ▶ integrate safety alarms.

Engineering Support

- ▶ gas-pipeline compatibility;
- ▶ electrical / battery readiness;
- ▶ preventive maintenance;
- ▶ leak and functional checks;
- ▶ ventilator performance verification;
- ▶ service and recall management.

Patient Monitoring in the OT

Typical monitored variables include:

- ▶ ECG and heart rate;
- ▶ non-invasive / invasive blood pressure;
- ▶ oxygen saturation;
- ▶ respiratory rate and capnography;
- ▶ temperature;
- ▶ anaesthetic-gas variables where applicable;
- ▶ additional specialty measurements for complex surgery.

System Challenge

Multiple devices must remain visible, audible and clinically interpretable without creating alarm overload or cable entanglement.

Electrosurgical Units and Surgical Smoke

Electrosurgery uses high-frequency electrical energy for cutting and coagulation.

Risks

- ▶ unintended burns;
- ▶ damaged insulation;
- ▶ poor return-electrode contact;
- ▶ interference with other devices;
- ▶ ignition in an oxygen-enriched field;
- ▶ surgical smoke exposure.

Controls

- ▶ correct accessory selection;
- ▶ cable / insulation inspection;
- ▶ appropriate return-electrode placement;
- ▶ smoke evacuation when indicated;
- ▶ staff training;
- ▶ preventive maintenance and testing.

Laparoscopic and Endoscopic Surgery Systems

A minimally invasive setup may combine:

- ▶ camera head and image processor;
- ▶ light source;
- ▶ monitor / display;
- ▶ insufflator and CO₂ supply;
- ▶ electrosurgical / energy device;
- ▶ suction-irrigation system;
- ▶ recording / integration equipment.

Biomedical Engineering Challenge

Several independent devices become one clinical system. Compatibility, connectors, displays, gas supply, power, software and cable routing must all work together.

Robotic and Hybrid Operation Theatres

Robotic OT

- ▶ larger equipment footprint;
- ▶ docking and movement envelope;
- ▶ network / software dependence;
- ▶ specialized instruments;
- ▶ emergency undocking plan.

Hybrid OT

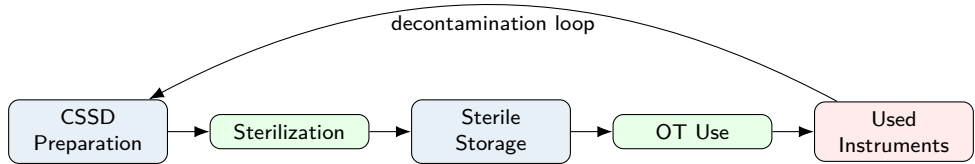
- ▶ surgical + advanced imaging capability;
- ▶ higher structural and electrical loads;
- ▶ radiation-safety requirements where applicable;
- ▶ complex ceiling / boom coordination;
- ▶ multidisciplinary commissioning.

Planning Lesson

Advanced technology increases the importance of early coordination between surgeons, anaesthesia, radiology, architects, IT and biomedical engineers.

Interface with CSSD

The OT depends on sterile instruments arriving in the right set, at the right time and in usable condition.



Sterile Instrument Flow and Traceability

A robust system should answer:

- ▶ Which instrument set was used for which patient?
- ▶ Was the sterilization cycle acceptable?
- ▶ Was the pack intact and within its accepted event / shelf criteria?
- ▶ Were implants and high-risk devices traceable?
- ▶ Did any instrument fail during the case?
- ▶ Was the complete set returned for decontamination?

Digital Opportunity

Barcode / RFID-based tray tracking can connect CSSD processing, OT use, instrument history and maintenance records.

Staff Flow, Attire and Access Control

- ▶ staff enter controlled zones through designated changing areas;
- ▶ access should be limited to personnel required for the procedure;
- ▶ unnecessary door opening should be minimized;
- ▶ traffic between contaminated and cleaner areas should be controlled;
- ▶ visitors, vendors and service personnel need defined access rules;
- ▶ emergency access must still be possible without disabling the control system.

NCDC Principle

People are a major source of airborne contamination in the OT; controlling traffic and door opening supports the intended clean-air environment.

Infection Prevention in the OT

Infection prevention combines engineering and behavior:

- ▶ appropriate hand hygiene and surgical preparation;
- ▶ sterile instruments and aseptic technique;
- ▶ controlled ventilation and pressure;
- ▶ environmental cleaning;
- ▶ appropriate attire and PPE;
- ▶ minimization of unnecessary traffic;
- ▶ correct handling of sharps and contaminated material;
- ▶ surveillance and investigation of surgical-site infections.

No single control can replace the complete infection-prevention system.

Cleaning and Turnover Between Cases

A safe turnover process may include:

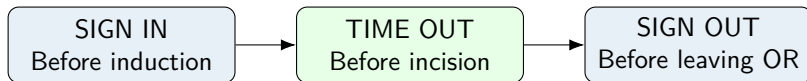
1. remove waste, sharps and used instruments through the defined pathway;
2. clean and disinfect patient-contact and frequently touched surfaces;
3. replace linen and consumables as required;
4. inspect visible contamination and spills;
5. restore equipment position and cable organization;
6. verify essential equipment readiness for the next case;
7. document / escalate any equipment or environmental problem.

Engineering Lesson

Turnover speed must never create a shortcut around cleaning, equipment checks or environmental safety.

WHO Surgical Safety Checklist

The WHO checklist organizes team checks around three critical moments:



Purpose

Standardized team communication helps prevent errors involving identity, site, equipment, anticipated critical events, specimens and postoperative planning.

Medication, Implant and Specimen Safety

Common OT risks include wrong drug, wrong concentration, wrong implant and wrong specimen identity.

- ▶ label prepared medications and solutions clearly;
- ▶ verify high-risk drugs and infusions according to policy;
- ▶ confirm implant identity, size and traceability before use;
- ▶ label specimens immediately and reconcile before dispatch;
- ▶ communicate special handling requirements to laboratory / pathology;
- ▶ document unused / returned implants and critical consumables.

Sharps and Biomedical Waste

At the Point of Use

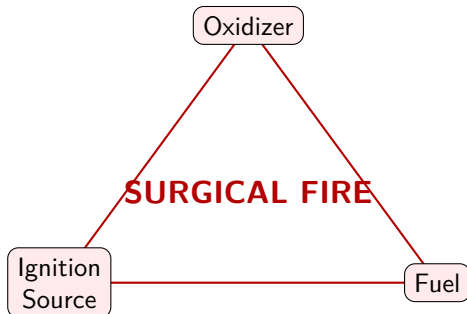
- ▶ safe passing technique;
- ▶ puncture-resistant sharps container;
- ▶ no unsafe recapping;
- ▶ immediate segregation;
- ▶ spill response readiness.

System Design

- ▶ bins positioned without obstructing staff;
- ▶ waste exits without crossing sterile supply;
- ▶ defined route to temporary storage;
- ▶ staff training and PPE;
- ▶ incident reporting for sharps injury.

Fire Safety in the Operation Theatre

The OT contains all elements needed for a potential surgical fire.



Examples: electrosurgery / laser + drapes / alcohol-based prep + oxygen-enriched atmosphere.

Preventing a Surgical Fire

- ▶ allow flammable skin-preparation agents to dry completely;
- ▶ control oxygen concentration and delivery near the surgical field when clinically feasible;
- ▶ keep ignition devices in safe mode when not in active use;
- ▶ inspect electrosurgical accessories and cables;
- ▶ communicate when a high-fire-risk situation exists;
- ▶ keep an appropriate emergency response plan and extinguisher access;
- ▶ train the team through drills and incident review.

Key Principle

Fire prevention requires coordination between anaesthesia, surgery, nursing and engineering – each team controls a different part of the fire triangle.

Equipment Safety, EMI and Cable Management

Multiple devices operating close together can create practical hazards.

- ▶ damaged cables and connectors;
- ▶ fluid ingress into electrical equipment;
- ▶ trip hazards from cords and hoses;
- ▶ electromagnetic interference between devices;
- ▶ blocked ventilation openings causing overheating;
- ▶ incompatible accessories or adapters;
- ▶ software / network failure in integrated systems.

Design Response

Use ceiling pendants / booms where suitable, adequate outlets, standardized accessories, cable-management routes and pre-use inspection.

Role of the Biomedical / Clinical Engineer

1. define technical requirements with clinicians;
2. review site readiness before procurement / installation;
3. coordinate equipment placement, power, gases, network and structural interfaces;
4. supervise installation, acceptance testing and commissioning;
5. maintain inventory, preventive maintenance and safety records;
6. investigate recurrent device failures and downtime;
7. support recalls, software updates and cybersecurity where relevant;
8. plan replacement and lifecycle cost for critical OT equipment.

Maintenance and Pre-use Readiness

System	Routine Check	Failure Consequence
Anaesthesia workstation	self-test, gas / circuit checks	ventilation / gas delivery failure
Patient monitor	sensors, battery, alarms	loss of physiological surveillance
Surgical light	movement, output, backup	inadequate surgical visualization
ESU	accessories, safety test, cables	burn / ineffective energy delivery
Suction	vacuum level, tubing, backup	inability to clear fluids / airway
HVAC	pressure / temperature alarms	loss of controlled environment

Site Readiness Before Installing OT Equipment

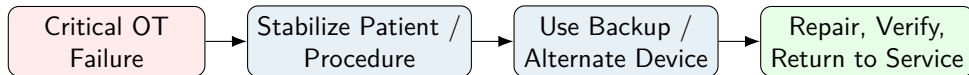
Ask before delivery:

- ▶ Is there enough usable space and movement clearance?
- ▶ Are floor / ceiling structural loads acceptable?
- ▶ Are electrical voltage, load, outlets and backup supply adequate?
- ▶ Are gas types, pressures and terminal locations correct?
- ▶ Is HVAC capacity sufficient for the equipment heat load?
- ▶ Are network, data, PACS / HIS interfaces needed?
- ▶ Can the equipment enter through doors, lifts and corridors?
- ▶ Is safe service access possible after installation?

Engineering Rule

A technically excellent device can still become a poor purchase if the room cannot support it.

Reliability and Downtime Planning



- ▶ identify devices and utilities whose loss can stop surgery;
- ▶ define backup equipment and manual alternatives;
- ▶ keep escalation contacts available 24/7;
- ▶ test emergency power, gas alarms and backup devices periodically;
- ▶ verify function after repair before clinical return.

Case Study 1: OT Loses Positive Pressure

During an elective procedure, the pressure monitor alarms and the room becomes neutral relative to the corridor.

Discussion Questions

1. Is the alarm genuine or a sensor fault?
2. Is the procedure already underway?
3. Have doors been repeatedly opened?
4. Is the AHU / fan operating normally?
5. Who decides whether the room remains suitable for continued use?
6. What must be checked before the next case?

Lesson

Environmental alarms need a clinical-engineering-infection-control response plan, not a simple reset.

Case Study 2: Medical Oxygen Pressure Drops

The central oxygen-pressure alarm activates while two operation rooms are in use.

Immediate Priorities

- ▶ verify alarm and source status;
- ▶ inform anaesthesia teams;
- ▶ prepare reserve cylinders / backup source;
- ▶ reduce non-essential demand if necessary;
- ▶ protect patients first.

Engineering Investigation

- ▶ source depletion / manifold issue;
- ▶ regulator failure;
- ▶ pipeline leak;
- ▶ valve position;
- ▶ alarm / sensor fault;
- ▶ maintenance / construction error.

Case Study 3: Unexplained Patient Burn

A patient develops a skin burn after a procedure in which electrosurgery was used.

Investigation Should Review

- ▶ return-electrode placement and contact;
- ▶ ESU settings and accessories;
- ▶ cable / insulation condition;
- ▶ contact with conductive objects or pooled fluids;
- ▶ device service and safety-test history;
- ▶ sequence of events and staff observations.

Lesson

Do not assume “operator error” or “device failure” before reconstructing the complete system and workflow.

Classroom Activity: Design a Two-OT Suite

Scenario

A 200-bed hospital plans two general operation theatres with a shared PACU and sterile support.

In groups, sketch a block layout and show:

1. outer, restricted, aseptic and disposal zones;
2. patient entry and postoperative exit;
3. clean instrument supply and dirty return route;
4. scrub / gowning and sterile storage;
5. PACU location;
6. major utility points;
7. one backup strategy for power or medical gases;
8. one infection-control risk you have intentionally reduced.

Mini-Project Ideas

- ▶ OT equipment downtime and root-cause dashboard;
- ▶ pressure / temperature monitoring and alarm-escalation system;
- ▶ site-readiness checklist for a robotic or laparoscopic OT;
- ▶ RFID / barcode-based surgical instrument traceability;
- ▶ FMEA of medical-gas failure in an operation theatre;
- ▶ surgical-light preventive-maintenance quality index;
- ▶ cable / boom layout redesign to reduce trip and access hazards;
- ▶ OT turnover-time study without compromising cleaning quality;
- ▶ energy-use assessment of OT HVAC during occupied and unoccupied periods.

Summary

- ▶ The OT suite is a controlled surgical system, not only an operating room.
- ▶ Zoning, access control, pressure relationships and circulation support contamination control.
- ▶ Surgical care depends on reliable HVAC, power, medical gases, suction, lighting, monitoring and anaesthesia systems.
- ▶ Sterile-supply and dirty-return pathways must be planned as complete loops.
- ▶ Fire, electrical, gas, equipment and infection risks require layered controls and team coordination.
- ▶ Biomedical engineers contribute from planning and procurement through commissioning, maintenance and incident investigation.

A safe operation theatre is engineered around the patient, the sterile field and the possibility of failure.

Assignment for the Next Lecture

Select one operation theatre in a hospital, or study a published / virtual OT layout, and prepare a **one-page engineering assessment**.

Include:

1. type of OT and major procedures performed;
2. a simple zoning / flow diagram;
3. five important biomedical devices;
4. three essential engineering utilities;
5. one infection-control design feature;
6. one safety-critical failure mode and backup strategy;
7. one practical improvement from a biomedical-engineering perspective.

Submission Format

One page, individually prepared, with a neat block diagram or layout sketch.

Suggested Reading and Safety References

- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ National Centre for Disease Control (NCDC), Ministry of Health and Family Welfare, *National Guidelines for Infection Prevention and Control in Healthcare Facilities*.
- ▶ World Health Organization, *WHO Surgical Safety Checklist* and implementation guidance.
- ▶ ANSI/ASHRAE/ASHE Standard 170-2025, *Ventilation of Health Care Facilities* – use the version adopted by the authority having jurisdiction.
- ▶ Relevant manufacturer manuals, hospital SOPs and local electrical / medical-gas / fire-safety requirements.

Next Lecture: Hospital Pharmacy Services

Thank You

Questions and Discussion